Conservation Genetics Practice Problems #5

1) What will the equilibrium allele frequencies be at a biallelic locus with a mutation rate of $\mu = 5 * 10^{-7}$ from A_1 to A_2 and $v = 1 * 10^{-8}$ from A_2 to A_1?

2) Assuming a per-base mutation rate of $1 * 10^{-}7$, what is the probability there will be no new single-nucleotide mutations in a new generation of 20 diploid organisms with 100,000 base genomes?

3) Solve for μ if the frequency of a recessive deleterious allele ($s = 0.1$) at mutation-selection equilibrium is 0.15.

4) Assuming a one-way input of migrants at rate m with a recessive allele at frequency q_m into a focal population where the allele is originally found at a frequency q_0, what will happen to the per-generation change in the frequency of q (i.e., Δq) over time?

   a) it will stay constant, because m does not change
   b) it will increase, as more copies of q are found in the population
   c) it will decline, as the frequencies of q in the migrant and focal popualtion become more similar
   d) there is not enouhg information to say

5) What evolutionary processes produce a cline in allele frequencies, and why?